Tezpur University

Spring End-Semester Examination-2024

Course Code: CO214 **Course Title: Computer Architecture and Organization**

Full marks=60 *Time= 3 hours*

(The figures in the right-hand margin indicate full marks for the questions)

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1. Choose the appropriate word to fill in the blank: **[1*7=7]**
   a. Cache memory is a form of __ (internal/external/non-volatile/optical) memory.
   b. In 1-address instructions, the second operand is implicitly taken from __ (stack/accumulator/MAR/IR)
   c. In __ (register/indexed/stack/relative) addressing mode, no memory fetch operation is needed for the operand.
   d. The registers, ALU, and internal CPU bus are collectively referred to as __ (processor/datapath/control unit/fetch unit)
   e. In __ (programmed IO/interrupt driven IO/direct memory access) the processor is involved throughout the entire data-transfer process from the main memory to I/O device.
   f. __ (PCI/SCSI/USB) is a high-bandwidth bus that allows devices to be connected to the system bus.
   g. __ (PCI/SCSI/USB) offers a plug and play feature.

2. State if the following statements are true or false, and justify your answer **(any four):** **[2*4=8]**
   a. The number of processor registers is an important instruction set design issue.
   b. PDP11 is a RISC processor.
   c. In the write-back policy of cache, the main memory data is always valid.
   d. DDR-SDRAM has a burst transfer mode for transferring a block of data.
   e. Virtual memory is part of the disk memory.

3. Consider the following C-code snippet and rewrite using PDP-11 instructions. Use registers R0, R1, and R2 for u, x, and y respectively. Registers R3 to R5 may be used for temporary data storage. **[5]**

```
u=0;
while(x >= y)
{   x = x - y;
    u++;
}
```

[Note: SUB A, B performs (A – B) → B; CMP A, B performs (A - B); MOV A, B performs A → B; INC A performs (A+1) → A; BNE/BLT/BLE performs branch if not equal/less than/less or equal]

4. Answer the following with appropriate diagrams **(any five):** **[4*5=20]**
   a. Explain *control hazard in pipelining* with a diagram. Give an example to show how reordering instructions may reduce the effect of delay slots.
   b. Explain the *handshaking protocol* of asynchronous data transfer. Draw the timing diagram for asynchronous read operation using handshaking protocol.
   c. Explain how *daisy chaining* is used for device identification in interrupt driven I/O.
   d. Explain *distributed bus-arbitration* using a diagram.
   e. Explain *fragmentation in variable partitioning* using a diagram.
   f. Explain *striping* in RAID devices using a diagram. Explain how it improves transfer time.

5. Consider a system with a 4GB main memory along with a cache memory of size 1MB. The system uses a word size of 1 byte and block size of 64B. For a 4-way set-associative cache, answer the following:
   a. Calculate number of addressable units, number of blocks in memory, and number of cache lines. **[3]**
   b. For the hexadecimal address AAAABBBB, show the tag, set, and word bits. **[2]**

6. Consider a system that uses a two-level cache. L1 cache access time is 5ns. If a word is in L2 cache but not in the L1 cache, 20ns are needed to load into L1 cache, and then the reference is started again for a hit. If the word is not in L1 nor in L2 cache but is in main memory, 50ns are required to fetch the word from memory to L2 cache, followed by 20ns to copy it to L1 cache, and then the reference is started again for a hit. The L1 cache hit ratio is 0.80 and the L2 cache hit ratio is 0.95. Calculate the average access time in this system. Compare the average access time if a single level cache with only the L2 cache were used. **[3+2]**

7. Consider a magnetic disk drive with 8 surfaces, 1024 tracks per surface, and 128 sectors per track. Sector size is 1KB. The average seek time is 6ms, the track-to-track access time is 1ms, and the drive rotates at 7200 rpm.

   a. Compute the size of the disk and the average rotational delay. **[2]**

   b. If a 5MB file is stored using sequential organization on a single platter, starting from Sector 0, Track 0, compute the number of sectors and tracks needed to store the file. If the current r/w head position is not known, compute the time required to read the entire file. **[3]**

8. Consider an empty cache memory having 4 lines that can hold 2 words each. If the following hexadecimal main memory block addresses are requested in the given order, then compute the number of cache hits and misses that will occur if an associative mapped cache with LRU policy is used.

   1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 10, 6, 7, 8, 9, 6, 7, 8, 9, 6, 10, 9, 10, 9, 7 **[5]**